PROJECT TRICORDER

MK I · FIELD OPERATIONS MANUAL

On-device multimodal instrument: chat, vision, image synthesis, neural speech, audio scene analysis, avian species ID, flora/fauna classification, append-only field log, and optional short-range camp link.

Field	Value
Software version	0.4.1 (versionCode 27)
Application ID	com.projecttricorder.app
Platform	Android 8.0+ (API 26), arm64-v8a
Reference hardware	Samsung Galaxy Z Fold3 (Snapdragon 888, 12 GB)
Document date	2 September 2026
Classification	Unclassified — operator manual
Companion artifacts	Tricorder-0.4.1-inquiry-fix.apk · project-tricorder-0.4.1.zip

1. Purpose and scope

This manual is the single operator and maintainer reference for Project Tricorder Mk I. It covers installation of the Android application and the four local inference services, day-to-day operation of every control, the theory of how data moves through the instrument, design rationale, a competitive feature matrix, and a fault tree.

It does not replace model licenses. Gemma, SmolVLM, Stable Diffusion, YAMNet, BirdNET, Kokoro, and Coral/iNaturalist weights remain under their upstream terms. The operator must accept those terms before downloading weights.

Tricorder is an instrument first and a chatbot second. The design goal is a pocket logger that can look, listen, speak, imagine, and write a timestamped book without a network account.

2. System overview

2.1 What the operator holds

Two cooperating halves:

- The APK — a Kotlin WebView shell with a phosphor CRT UI, native speech recognition, YAMNet and Task Vision classifiers in-process, a field log, and Google Nearby Connections.
- Termux services — four localhost HTTP servers the APK treats as OpenAI-compatible backends: Gemma chat on 8080, SmolVLM vision on 8081, Stable Diffusion Turbo on 8188, Kokoro TTS on 5000.

The APK will open and show sensor chrome without Termux. Inquiry, live vision captions, image generation, and spoken replies require the matching service to be up. Audio scene analysis, BirdNET, and flora/fauna classification do not require Termux; they run inside the APK.

2.2 Operating modes

Control	Mode	Primary path	Side effects
I Inquiry	CHATTING	Mic or type → Gemma :8080 → Kokoro :5000	IMAGE_GEN keywords route to SD :8188
II Sensor	SCANNING	Camera JPEG → SmolVLM :8081 + flora TFLite	Auto-hold after caption; writes field log + snap
III Sentry	GUARDING	Motion / jerk threshold → vision caption	Alert, klaxon, field log + snap
IV Audio	LISTENING	Duty-cycle mic → YAMNet → optional BirdNET	JSONL + field log; optional camp-link row
H Hold	Latch	Freezes loop; classifies current frame	Flora first, then SmolVLM
C Continuous	LIVE VISION	Sensor loop without Kokoro	Screen only; does not flood TTS
L Log review	BOOK	Slider over field.jsonl	Shows text + JPEG snap
E Log entry	OFFICER	Dictation stored, not sent to Gemma	kind=officer
V Video log	REC	Front camera MediaRecorder	Best-effort; may fail if WebView owns camera

3. Architecture

3.1 Process map

The APK process (com.projecttricorder.app) hosts WebView + JS tuple engine + Kotlin bridges. Termux is a separate Linux userspace. They meet only on 127.0.0.1. No cloud endpoint is required for the four backends.

Layer	Implementation	Notes
Presentation	WebView + tricorder-ui-shell.html	CRT, matrix, sliders, lights
Intent / routing	JS TupleEngine	button seed + keyword hints → route_to
JS ↔ native	TricorderBridge @JavascriptInterface	mic, Kokoro POST, audio sense, flora, field log, Nearby
On-device classifiers	TFLite Task Audio + Task Vision 0.4.4	YAMNet in APK; BirdNET and Coral packs downloaded to filesDir
Chat LLM	llama-server :8080	Gemma 3 4B Instruct Q4_K_M
Vision VLM	llama-server :8081	SmolVLM2 500M Video-Instruct Q8 + mmproj f16
Image gen	sd-server :8188	SD 2.1 Turbo Q4, cfg-scale 1, operator sliders
Speech out	Kokoro ONNX :5000	PocketPal-class graph + voices-v1.0.bin
Speech in	Android SpeechRecognizer	On-device STT; no Google cloud required if offline pack present
Book	FieldLog + AudioLog JSONL	files/logs/field.jsonl + snaps/*.jpg
Camp link	Play Services Nearby P2P_STAR	JSON rows only; radio range, not internet

3.2 Local ports

Service	Bind	Health
chat / Gemma	127.0.0.1:8080	GET /health or /v1/models
vision / SmolVLM	127.0.0.1:8081	GET /health
image / SD Turbo	127.0.0.1:8188	root banner or generations POST
tts / Kokoro	127.0.0.1:5000	GET /health
Termux sshd (optional)	0.0.0.0:8022	ssh from a desk

All four inference sockets bind loopback on purpose. They are not exposed to the LAN. The APK is the only client.

3.3 Storage map (on device)

Path	Contents
/data/data/com.projecttricorder.app/files/models/	BirdNET + Coral plant/insect/bird .tflite + labels

/data/data/com.projecttricorder.app/files/logs/field.jsonl	Unified book: audio, bird, flora, sensor, sentry, officer, video
/data/data/com.projecttricorder.app/files/logs/snaps/	JPEGs referenced by field.jsonl snap field
/data/data/com.projecttricorder.app/files/logs/audio.jsonl	Legacy per-listen YAMNet/BirdNET rows
~/models/ (Termux)	Gemma, SmolVLM, mmproj, SD Turbo GGUF
~/kokoro/	server.py, kokoro ONNX, voices-v1.0.bin
~/tricorder-logs/	chat.log vision.log image.log tts.log

4. Theory of operations

4.1 Tuple routing

Each hardware button carries a four-tuple seed (mode, input channel, target modality, action). Optional keyword hints rewrite target modality before a rule table selects a backend. Example: Inquiry seed is CHATTING / AUDIO_IN / TEXT / TOGGLE. If the utterance contains “show me” or “imagine”, target becomes IMAGE_GEN and the request is posted to sd-server instead of Gemma.

The wake word “computer” is intercepted in JS before the model. It plays a scripted ship’s-computer monologue (service times, sensors, earth-link ping) and does not consume a Gemma turn unless residual text remains.

4.2 Inquiry path

1. Operator taps I. CRT shows Awaiting query.
2. Mic uses Android SpeechRecognizer. Partials fill the box; the final result calls submitInquiry().
3. If Log entry (E) is armed, the line is written as kind=officer and Gemma is not called.
4. Otherwise the tuple engine picks chat_vision or image_generation.
5. Gemma returns text. The APK POSTs that text to Kokoro and plays the WAV with MediaPlayer.
6. Backend lamps: green idle, amber in-flight, red unreachable. They poll /health.

4.3 Sensor path

SCANNING runs on a loop (default 12 s). Each tick captures a 320×240 JPEG data URL, posts it to SmolVLM with a short describe prompt, and simultaneously runs the same JPEG through three Coral ImageClassifiers (plant, insect, bird). The CRT concatenates the caption and FLORA/FAUNA lines. The book stores caption + flora + snap + UTC + optional GPS. After one caption the UI latches Hold so Kokoro is not flooded.

4.4 Sentry path

GUARDING polls a coarse camera delta and the device accelerometer jerk. If either crosses its threshold and the cooldown has expired, one frame is sent to SmolVLM, an alert is raised, Kokoro speaks unless Continuous is on, and the book stores the alert plus snap.

4.5 Audio path

YAMNet lives in the APK (Task Audio). A background thread records PCM for dwell seconds (slider, 1–20 s), classifies overlapping 1 s windows, sleeps for interval seconds (slider, 5–60 s), and posts JSON to the CRT. If a top class contains Bird or Animal and the BirdNET file is present, the same PCM is classified for species. Waveform bins are peak envelopes, not a spectrogram. This path does not use Termux.

4.6 Flora path

Coral iNaturalist MobileNet-v2 224 quantized graphs are ImageClassifier models with embedded labels. The APK must use TFLite Task Vision — a raw Interpreter ignores metadata and returns empty hits. Packs are fetched at runtime into the app private models directory, not Downloads. Classification is in-process and does not need 8081, though Sensor still uses 8081 for the English caption.

4.7 Camp link path

Google Nearby Connections, strategy P2P_STAR, service id com.projecttricorder.hits. Both devices advertise and discover. On connect, JSON rows may be sent to peers. In 0.4.1 the only automatic radio payload is the audio-listen row. Sentry and flora rows stay local unless a later build forwards field.jsonl kinds. Radio range is metres, not kilometres. Location, Nearby devices, Bluetooth, and Wi-Fi must be granted or discovery fails.

5. Capabilities and limits

Capability	Works offline	Typical cost on Fold3	Quality note
Gemma 4B chat	Yes, if service up	Tens of seconds per reply	Instruction-tuned; not a 70B
SmolVLM caption	Yes	A few seconds per frame	Short, concrete; not a taxonomist
SD Turbo stills	Yes	~30 s at 384–512 / 4–6 steps	Unfinished below 4 steps; sliders exist
Kokoro speech	Yes	On the order of PocketPal	af_bella; strip periods in long readouts
YAMNet scene	Yes (in APK)	Per dwell window	521 AudioSet classes
BirdNET species	Yes after fetch	When YAMNet says bird	Needs the downloaded tflite
Coral flora/fauna	Yes after fetch	On each Sensor/Hold frame	iNaturalist labels; wrong-pack noise at low %
Field book	Yes	Negligible	JSONL + JPEG; not a cloud sync
Camp link	No internet needed	Radio pairing time	Nearby range only; 0.4.1 sends audio rows
Officer / video log	Yes	Video is best-effort	E steals Inquiry line by design

What it is not: a replacement for a licensed field guide, a long-range mesh radio, a medical device, or an always-correct species oracle. Low-confidence second packs (plant net naming a turkey on a shrub) are expected. Treat percentages as scores, not probabilities of truth.

6. Installation

6.1 Sideload the APK

1. Copy Tricorder-0.4.1-inquiry-fix.apk to the phone (USB, Drive, or Downloads).
2. Settings → Security → allow install from that source.
3. Install. Open once. Grant Camera, Microphone, Location. Optionally Nearby devices / Bluetooth.
4. Battery: set Termux (later) and Tricorder to Unrestricted so duty cycles survive Doze.

Minimum: 64-bit ARM Android 8. 8 GB RAM is tight once four servers are up; 12 GB is the reference. Uninstall older 0.3.x builds before 0.4.1 if WebView assets look stale.

6.2 Termux and compilers

Install Termux from F-Droid, not the abandoned Play listing.

1. pkg update && pkg install wget curl git clang cmake python openssh termux-api espeak

2. termux-setup-storage && termux-wake-lock && termux-change-repo
3. Copy termux/tricorder from the project zip to ~/bin/tricorder and chmod +x.

6.3 Weights

Place GGUF files in ~/models/ :

- gemma-3-4b-it-Q4_K_M.gguf — accept the Gemma license on Hugging Face first or llama-server -hf will 401.
- SmolVLM2-500M-Video-Instruct-Q8_0.gguf and mmproj-SmolVLM2-500M-Video-Instruct-f16.gguf.
- stable-diffusion-v2-1-turbo-Q4_0.gguf (or the turbo checkpoint you built against).

fetch-models.sh in the zip is resume-safe. Kokoro: copy ~/kokoro including the ONNX graph used with PocketPal and voices-v1.0.bin. On this ORT/Android combination, fp16 produced NaNs; keep the working graph that already speaks “Tricorder online.”

6.4 Binaries

aarch64 Termux ELFs copy between Termux installs of similar vintage. They do not run on stock Android. If llama-server dies with CANNOT LINK EXECUTABLE, rebuild:

- llama.cpp: cmake -B build && cmake --build build -j --target llama-server
- stable-diffusion.cpp: cmake -B build -DSD_WEBP=OFF && cmake --build build -j --target sd-server

SD_WEBP=OFF avoids the missing Android cpu-features / libwebp hole.

6.5 Start the stack

First start from the Termux app UI, not from an SSH session that can die.

- ~/bin/tricorder start
- Wait 10–20 s, then ~/bin/tricorder status

Expect four lines up: chat :8080, vision :8081, image :8188, tts :5000. sshd on 8022 is started as a convenience and is left running on stop.

6.6 In-APK model fetches

Open backend & camera. Fetch BirdNET (~25 MB) and Fetch flora/fauna (~12 MB). Progress appears on the CRT. Files land in the app private models directory. Desktop downloads of the same URLs are not visible to the APK.

7. Guide to operations

7.1 Lights

- IDLE / ACTIVE / ALERT — instrument state.
- CHAT VISION IMAGE VOICE — backend reachability. Green reachable, amber in-flight, red down.
- Wait for green before stacking a new spoken query on Kokoro.

7.2 Inquiry

Tap I. Type or Mic. Speak naturally. “Show me the planet Vulcan” routes to image generation. “Computer” alone plays the monologue. After 0.4.1 the box clears and the CRT shows Working… while Gemma runs. If E Log entry is red/engaged, your sentence is a log line, not a question.

7.3 Sensor

Tap II. Point at the subject. One cycle produces an English caption plus flora/fauna scores. Hold turns red after the cycle. Tap Hold again (green) to resume scanning. Continuous writes captions without Kokoro. For a clean plant ID, fill the frame with the organism; the 224-net is not a detector, it classifies the whole JPEG.

7.4 Sentry

Tap III. Set the device with a view of the scene. Motion or a bump raises an alert, speaks unless Continuous is on, and writes the book. Do not leave it aimed at a television if you do not want frame-change alerts.

7.5 Audio

Tap IV. Set listen / sleep sliders in backend (example: listen 5 s, sleep 15 s). Scene labels appear with a waveform. BirdNET runs only when YAMNet reports Bird or Animal. Hold pauses the duty cycle.

7.6 Image sliders

Backend panel: size 256 / 384 / 512 and step count. Four steps at 512 recovered coherent cats on the Fold3; two steps produced color blocks. Treat the sliders as a quality/time trade, not a hidden GPU switch. This build is CPU-side sd-server.

7.7 Log review and officer entry

L opens the slider. Each notch is one field.jsonl row: kind, local time, text, snap if present. E arms officer mode. V starts a front-camera clip; tap again to stop. If V reports failure, the WebView camera session already owns the sensor — leave Sensor first.

7.8 Camp link

1. Same APK on both phones. Location + Nearby + Bluetooth + Wi-Fi granted.
2. backend → Camp link on, on both units.
3. Wait for LINK advertising / discovering / linked.
4. Leave camp unit in Audio or Sentry. Carry the other within radio range.

0.4.1 forwards audio-listen JSON. Status strings appear as LINK … on the CRT. Restore of the remote-hit painter and forwarding of sentry/flora kinds is a subsequent software change, not a field setting.

8. Design notes

- Split brain: heavy generative models stay in Termux so the APK can be rebuilt without relinking llama.cpp. Classifiers that must survive a dead 8081 live in-process.
- Loopback only: generative servers never bind 0.0.0.0.

- Hold is a flood valve. Sensor and Sentry latch so Kokoro is not stacked.
- Task APIs over raw Interpreter for YAMNet and Coral — metadata carries labels and normalization.
- Field log is append-only JSONL. There is no edit UI. That is intentional.
- Star Trek chrome is skin. The book, duty cycle, and local models are the product.
- FP16 Kokoro NaN'd on this ORT build; the working PocketPal-class graph is the supported one.

9. Competitive matrix

Comparison is of shipped operator-visible capability on a phone, not of research demos. "Y" means the feature is usable offline in a current public build without a vendor account. "P" means partial, plugin, or cloud fallback. "—" means not in product scope. Assessments are conservative as of September 2026.

Capability	Tricorder 0.4.1	PocketPal AI	AI Edge Gallery / Gemini Nano	Apple Intelligence	llama.cpp UIs (ChatterUI & kin)	Cloud assistants
Offline chat LLM	Y	Y	Y	P	Y	—
Offline vision VLM	Y	P	P	P	P	—
Offline image generation	Y	—	—	—	—	P
Neural TTS on device	Y	Y	P	Y	P	P
On-device STT	Y	P	P	Y	P	P
Audio scene classifier	Y	—	—	—	—	—
Bird species ID	Y	—	—	—	—	P
Flora/fauna vision net	Y	—	—	P	—	P
Unified field log + snaps	Y	—	—	—	—	P
Duty-cycle sentry	Y	—	—	—	—	—
P2P camp alert link	P	—	—	—	—	—
No account required	Y	Y	P	—	Y	—
Open weights operator-owned	Y	Y	P	—	Y	—
Single instrument UI	Y	—	—	P	—	—
Concurrent local backends	Y	—	—	—	—	—

Reading the matrix: PocketPal is the closest peer for chat + Kokoro-class TTS and is the source of the working ONNX graph. It is not an instrument. Google's on-device gallery and Apple Intelligence integrate with the OS and often fall back to cloud for multimodal work. llama.cpp frontends expose models; they do not run a sentry loop or write a geotagged book. Cloud assistants win on quality and lose on air-gap, account, and radio silence.

Tricorder's distinctive cell is the conjunction: four local generative services + in-APK bioacoustics and taxa nets + an append-only log + a physical control surface. Individual cells exist elsewhere. The conjunction does not.

10. Troubleshooting

10.1 Inquiry stuck on Awaiting query

Fixed in 0.4.1. Older shells called isAudioLogQuestion without defining it; submit threw and the box never cleared. Install 0.4.1. If it returns on 0.4.1, confirm I is engaged (not E Log entry) and CHAT is green.

10.2 Audio armed, waveform, no labels

0.3.3–0.4.0 HTML called fmtScore without defining it. 0.3.2 and 0.4.1 use Number(score)*100. Uninstall the broken build; do not keep two WebView asset generations.

10.3 Flora never appears

- Fetch flora/fauna inside the APK and wait for the CRT note. Desktop files in Downloads are invisible.
- Sensor must deliver a JPEG. Live Sensor in 0.3.7+ passes that JPEG into classifyFlora. Hold-only builds before that waited on SmolVLM and looked empty.
- If the line is “pack not on device”, the tflite is missing from files/models.
- If plant% is low and a wrong pack names a turkey, ignore sub-35% foreign packs.

10.4 Vision lamp red / HOLD eval failed fetch

SmolVLM is down. ~/bin/tricorder status. If vision is down: ~/bin/tricorder logs vision. Common causes: missing mmproj, OOM after Gemma already holds RAM, SSH session died and took nohup with it (always start from the Termux app). Restart: ~/bin/tricorder restart.

10.5 Image 400 or doubled/low-res garbage

- 400: APK posted a body sd-server rejected. Confirm :8188 is sd-server, not another process.
- Color squares: steps too low. Use 4 at 512 or 6 at 384.
- Minutes per 256 px: CPU path plus thermal throttle. Lower steps, not only resolution.
- Do not expect Adreno offload in this build.

10.6 Kokoro silent or NaN WAV

- VOICE red: :5000 down. Check ~/tricorder-logs/tts.log.
- Silent 94 KB WAV: bad style rank or fp16 graph. Use the PocketPal-class graph that already spoke.
- Drops words on “U.S.S.” and decimals: strip periods before POST (already done for the computer monologue).
- Flooded TTS: turn Continuous on, or leave Hold latched.

10.7 llama-server 401 / invalid username

Hugging Face gated Gemma. Log in on a browser, accept the license, retry, or pass a token. Local -m ~/models/….gguf avoids -hf after the file exists.

10.8 Camp link never says linked

- Location must be on. Nearby devices + Bluetooth + Wi-Fi granted.
- Toggle Camp link off/on both phones. Pair at one table first.
- Play Services present. AOSP without GMS will not run this Nearby client.
- adv fail / disc fail is almost always permission or GMS, not the SERVICE string.

10.9 Video log fails

Legacy Camera API plus WebView getUserMedia contend for the same sensor. Leave II/III, tap V. If it still fails, officer audio log is the reliable record.

10.10 Termux services die after desk SSH drops

Start with ~/bin/tricorder start from the phone. The script uses setsid nohup. Do not rely on a laptop SSH TTY as the parent. status/logs work over SSH afterward.

10.11 Thermal / RAM

Four servers plus Sensor plus SD will heat a Fold3. If vision or chat is killed, drop SD first, then cut Sensor loop rate. 8 GB devices should run chat+TTS only.

11. Maintenance

- Update APK by uninstall/install when WebView assets change (0.3.x → 0.4.x).
- Do not delete files/models unless you intend to Fetch again.
- field.jsonl grows forever. Copy off device with adb exec-out run-as when you want an archive.
- Model licenses remain upstream. Do not redistribute Gemma weights inside the zip.

12. Document control

Rev 0.4.1 describes software version 0.4.1 as built 1 September 2026 and operated 2 September 2026. Features marked partial (camp-link payload set, video log) are documented as they exist, not as a roadmap. Future revisions should bump both the APK versionName and this Rev together.